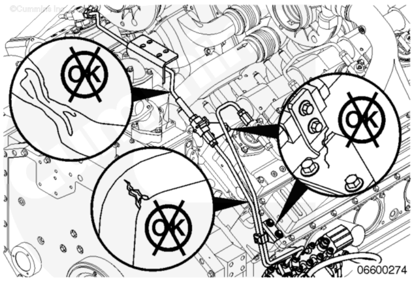
## Trainings

## Initial Check

with Electronically Actuated Injector

**⚠ WARNING ⚠**

**The fuel pump, high-pressure fuel lines, and injector supply lines contain very high-pressure fuel. To reduce the possibility of injury and property damage, do not loosen any fittings while the engine is running. Carefully follow the following steps to properly relieve the pressure.**



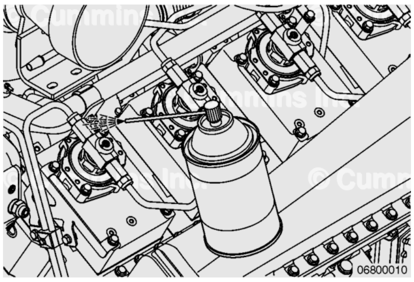

Inspect the injector high-pressure supply lines for cracks, chafing, leaks, and loose or broken clamps.

## Preparatory Steps

with Electronically Actuated Injector

**⚠ CAUTION ⚠**

**A small amount of dirt and debris can be very harmful to the injectors and the cone seats on the injector high-pressure supply connections. Extra care is required to keep the fuel connections clean during removal and installation. See Fuel System heading in Procedure 204-008 (General Cleaning Instructions).**




- Clean the engine. Refer to Procedure 000-009 in Section 0. (/qs3/pubsys2/xml/en/procedures/28/28-000-009.html) Refer to Procedure and 204-008 in Section i. (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
- Use contact cleaner, Part Number 3824510, or equivalent, to thoroughly clean all fuel lines before removal from the engine. Clean the connector fittings to remove as much debris as possible. It is very important that extra care is

taken to keep the fuel connections clean during removal and installation. Refer to Procedure 204-008 in Section i. (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)

- Make sure that debris, water, steam, or cleaning solution does **not** get inside the fuel system.

## Remove

### with Electronically Actuated Injector

**⚠ WARNING ⚠**

**Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.**



**⚠ WARNING ⚠**

**The pressure of the fuel in the line can penetrate the skin and cause serious personal injury. Wear gloves and protective clothing when working on the fuel system.**

**⚠ WARNING ⚠**

**Pressure within the high-pressure fuel system must never be measured using a mechanical gauge. Fuel pressure values of over 1724 bar [25,000 psi] are possible. Under high pressure, a mechanical gauge can fail, causing a high-pressure fuel leak resulting in personal injury and property damage.**

**⚠ CAUTION ⚠**

**A small amount of dirt and debris can be very harmful to the injectors and the cone seats on the injector high-pressure supply connections. Extra care is required to keep the fuel connections clean during removal and installation. See Fuel System heading in Procedure 204-008 (General Cleaning Instructions).**

**⚠ CAUTION ⚠**

**Do not overtighten or undertighten the injector supply line fittings. A fuel leak can occur.**

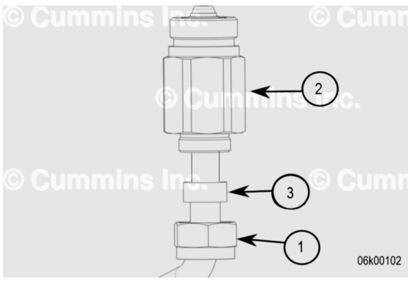
**⚠ CAUTION ⚠**

**Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.**

**Note :** The pressure from the fuel system **must** be relieved before removing the rail pressure sensor.

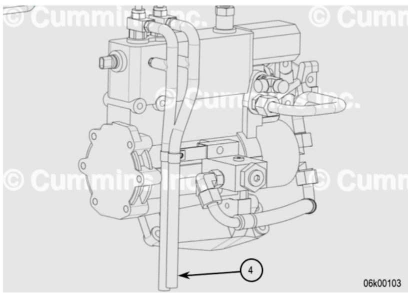
**Note :** The drain line **must** be checked for blockage before relieving the pressure to prevent fuel line rupture.

Loosen the brass nut (1) that clamps the rubber grommet to the high pressure nut at the end of the injector. Put a M27 [1-1/16 in] wrench on the high pressure nut (2) next to the injector to prevent it loosening. Slide back the rubber grommet (3) if the rubber grommet is found to be or degraded it must be discarded and replaced.

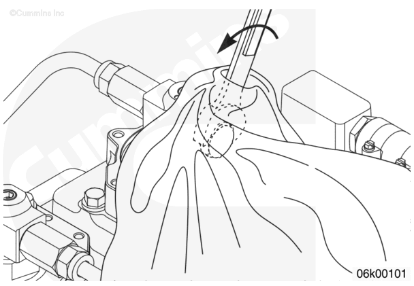



Apply compressed air (max 150psi), into the rubber drain line closest to the fuel pump. (4) The hose closest to the front of the engine connects to the right bank, the hose closest to the rear connects the left bank. If air is released in between the high-pressure nut and pipe, then the drain line is clear. **If air is not released, then contact your local Cummins Inc. distributor for assistance.** If the lines are clear re-fit the rubber grommet (3) and brass nut (1). See graphic in the stepblock above.

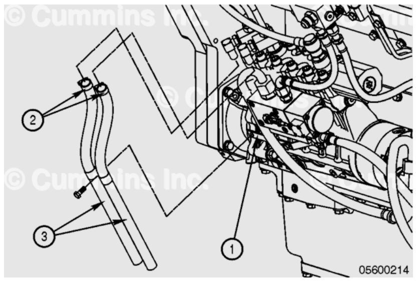
**Torque Value:** 45 n•m [ 33 ft-lb ]




Cover the plug and adjoining joint with a lint-free cloth. Slowly loosen the plug on the last injector of the high-pressure fuel lines ¼ to ½ turn. Stay out of the path of potential fuel spray. The plug does **not** need to be removed to relieve the pressure.



Connect INSITE™ electronic service tool and verify the fuel pressure has bled down by monitoring the fuel rail pressure.

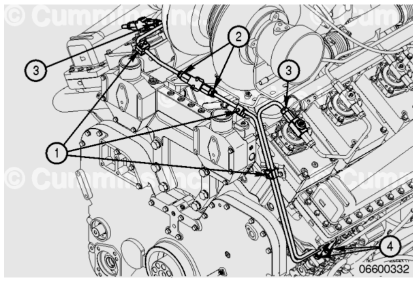



Remove the capscrews and P-clips that secure both the fuel vent hoses to the supporting bracket (1). (Other applications may have a variation to the fuel drain hardware.)

Loosen the two clamps securing the fuel vent hoses (2).

Remove the hoses (3).

Disconnect the three clamps (1) securing the injector supply lines to the engine.

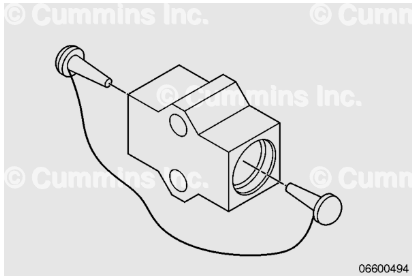



Loosen the rectangular grommet nut from the connection block (2) and the supply lines to the first injector on the left and right banks (3). Support the connector nut with a wrench while loosening the rectangular grommet nut. Slide the rubber grommets from the connecting fittings.

Disconnect the injector supply lines from the fuel pump (4).

Remove the injector supply line from the engine and discard the o-rings.

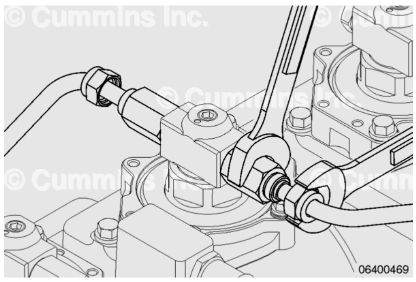
Do **not** reuse the protective plugs. The protective plugs **must** be used immediately upon removal from the plastic wrapping. If they are **not** used immediately, the protective plugs **must** be discarded. If the protective plugs are fouled in any manner before use, they **must** be discarded and a new set used.




Plug the ports in each side of the connection block with protective plugs, Part Number 4918767.

The plugs **must** be firmly wedged in place to prevent debris from entering.

**Note : On QSK38 and QSK50 engines, all the V-shaped bends in the injector supply lines are oriented in the downward direction.**

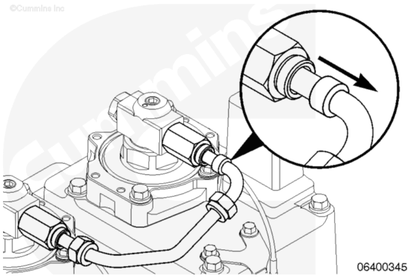



**Note : On QSK50 Marine application engines, the V-shaped bends in the injector supply lines are alternatively oriented in the downward and upward direction.**

Loosen the rectangular grommet nuts from each side of the injectors.

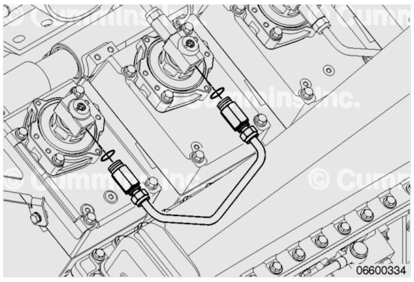
Support the connector nut with a wrench while loosening the rectangular grommet nut.

Slide the rubber grommets away from the connecting fittings.




Loosen the connector nuts and remove the injector supply lines between the injectors from the engine.

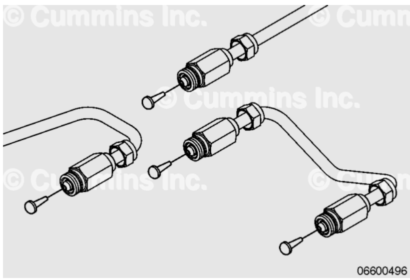
Remove and discard the o-rings.




Do **not** reuse the protective plugs. The protective plugs **must** be used immediately upon removal from the plastic wrapping. If they are **not** used immediately, the protective plugs **must** be discarded. If the protective plugs are fouled in any manner before use, they **must** be discarded and a new set used.

Immediately after removing the injector supply line, plug each end of the line with protective plugs.

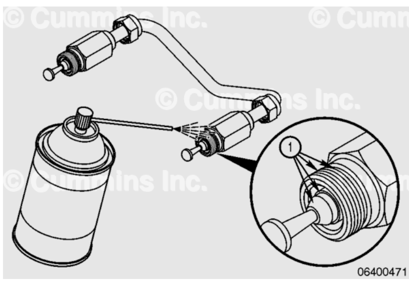
The plugs **must** be firmly wedged in place to prevent debris from entering.




# Clean and Inspect for Reuse

## with Electronically Actuated Injector

Thoroughly clean the ends of the injector supply line with contact cleaner, Part Number 3824510, or equivalent. The target areas for cleaning include the connector fitting threads, o-ring groove, sealing cone, and inside the connector fitting end (1).

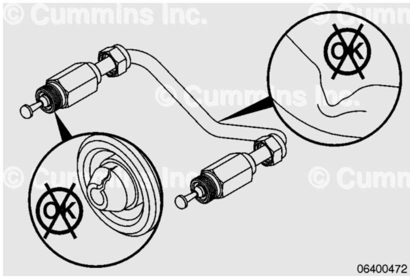


Clean the target areas to remove as much debris as possible, by spraying a sufficient amount of contact cleaner (approximately 5 seconds).

Make sure the protective plug does **not** fall out as the contact cleaner is sprayed on the end of the injector supply tube line.

If any visible debris can **not** be removed from the target areas (1) that may cause harmful contamination to the fuel system after installation, discard the injector supply line and replace as necessary.

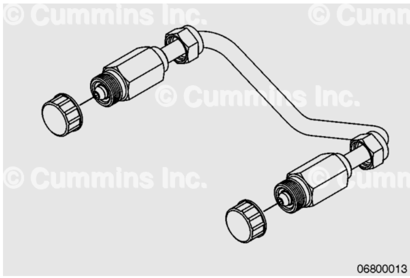
Inspect the end of the injector supply line for damaged surfaces, especially at the sealing cone.




If there are signs of burrs, material rounding, or cracking, replace as necessary.

Check for cracks, wear, or pinched areas along the injector supply line. Replace as necessary.

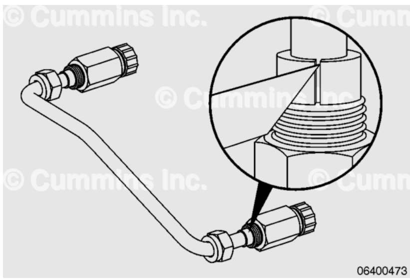
Do **not** reuse the yellow threaded caps. The yellow threaded caps **must** be used immediately upon removal from the plastic wrapping. If they are **not** used immediately, the yellow threaded caps **must** be discarded. If the yellow threaded caps are fouled in any manner before use, they **must** be discarded and a new set used.




Remove the protective plugs and cap the ends of the injector supply line with the yellow threaded caps to prevent debris from entering the high-pressure line.

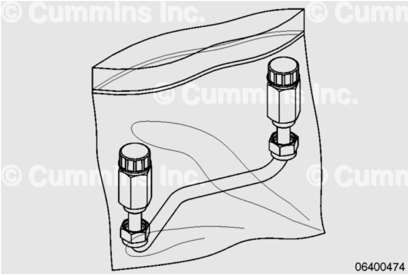
**⚠ CAUTION ⚠**

**Make sure the injector supply line is not damaged as the rubber grommets are sliced off.**




Remove the rubber grommets by slicing the grommets off with a sharp razor. Discard the rubber grommets.

Bag the fuel line in a clean bag and seal the bag. Use one clean bag per fuel line.



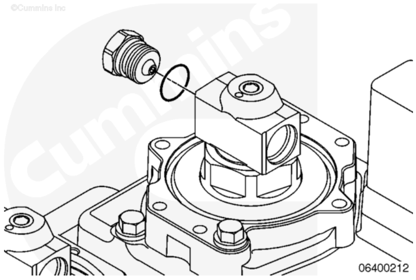

Remove the plug on the last injector on each bank and discard the o-ring.

Spray the plug with contact cleaner, Part Number 3824510, or equivalent.

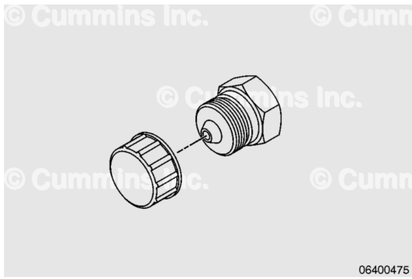
If any visible debris can **not** be removed from the plug, especially the sealing cone, discard the plug and use a new plug.

Check the plug for damage at the sealing surfaces.

If there are any signs of burrs, material rounding, or cracking, replace as necessary.



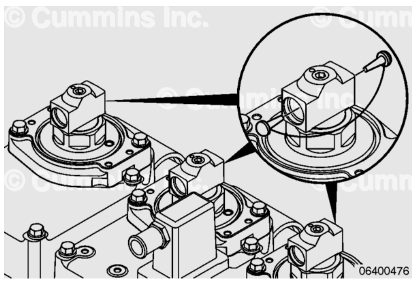

Cap the plug with a yellow threaded cap.




Do **not** reuse the protective plugs. The protective plugs **must** be used immediately upon removal from the plastic wrapping. If they are **not** used immediately, the protective plugs **must** be discarded. If the protective plugs are fouled in any manner before use, they **must** be discarded and a new set used.

Plug the ports in each side of the injector tee-piece with protective plugs, Part Number 4918767.

Make sure the plugs are wedged firmly in place to prevent debris from entering and causing damage to the injector body.

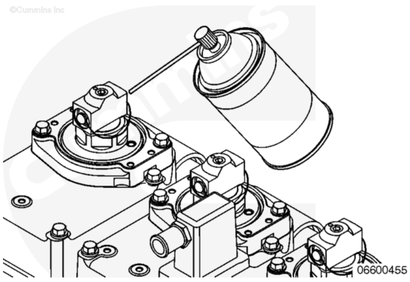


# Install

## with Electronically Actuated Injector

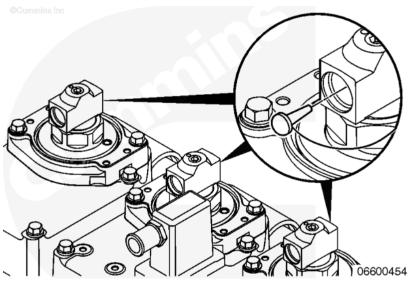
**⚠ CAUTION ⚠**

**A small amount of dirt and debris can be very harmful to the injectors and the cone seats on the injector high-pressure supply connections. Extra care is required to keep the fuel connections clean during removal and installation. See Fuel System heading in Procedure 204-008 (General Cleaning Instructions).**



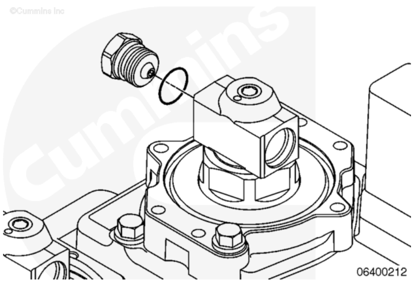

With the injector protective plugs still installed on the injector ports, clean the injector ports with contact cleaner, Part Number 3824510, or equivalent. Make sure the plugs remain firmly wedged in place.

Remove the injector protective plugs from the ports in the injector T-fitting and discard the plugs.




**⚠ CAUTION ⚠**

**Do not overtighten or undertighten the injector supply line fittings. A fuel leak can occur.**

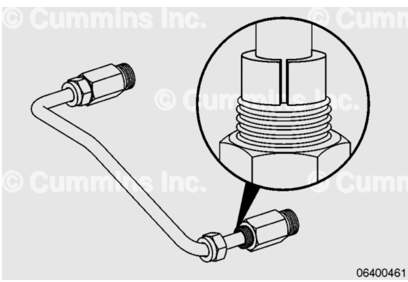



Assemble a new o-ring onto the plug.

Install the plug into the last injector connection at each bank of the engine.

**Torque Value:** 45 n•m [ 33 ft-lb ]

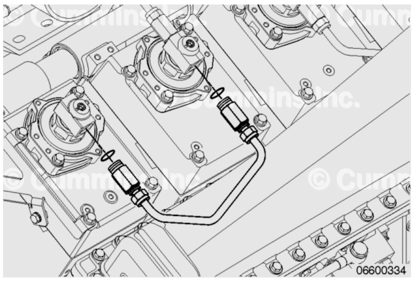
Remove the injector supply line from the clean bag.




Lubricate the pre-sliced grommets with clean engine oil on the outside and inside diameters and install on the outer wall of the injector supply line.

Remove the yellow threaded caps from the ends of the injector supply line and discard the caps.

Install new o-rings on each cone end of the new injector supply line.




**Note : On QSK38 and QSK50 engines, all the V-shaped bends in the injector supply lines are oriented in the downward direction.**

**Note : On QSK50 Marine application engines, the V-shaped bends in the injector supply lines are alternatively oriented in the downward and upward direction.**

It is important to install both ends of the new injector supply line simultaneously to each pair of injectors. If one end of the fuel line is installed and tightened before the other, the fuel line will have to bend in order to install and tighten the other end.

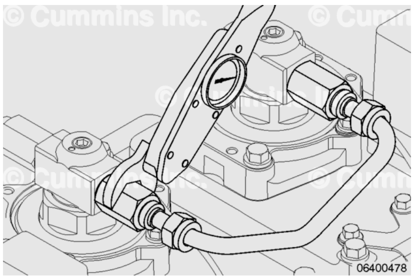
Insert the cone seat tips of both ends of the injector supply line into the injectors. Be careful **not** to damage the cone seats during installation.

Start the first thread of the connector nut on both ends of the injector supply line by hand. **Only** start the first thread for one connector, then start the first thread for the second connector. Be careful to properly engage the first thread to reduce the possibility of cross-threading. If the thread can **not** be started by hand, the injector clamp needs to be loosened and the injectors rotated until the injector supply line can be started by hand. Refer to Procedure 006-026 (Injector) in Section 6 for removal, installation, and alignment of the injectors. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)

Thread the connector nuts into the injectors until finger tight. Both nuts **must** be snug before tightening either end.

Check for alignment or installation issues before final tightening.

Tighten both connector nuts with a suitable M27 [1-1/16 in] crow's foot and torque wrench.

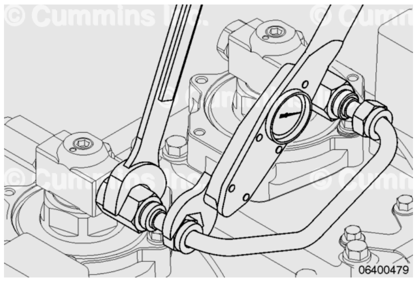



**Torque Value:** 45 n•m [ 33 ft-lb ]

Install the rectangular grommet nuts onto the connector nuts at the connection block and the first injector of each bank with a suitable M24 [15/16 in] crow's foot and torque wrench.

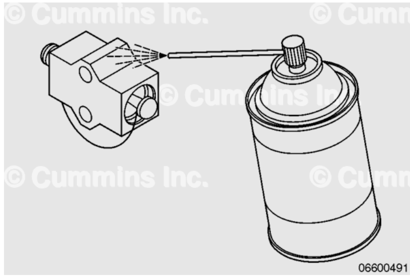
Support the connector nut with a wrench while tightening the rectangular grommet nut.

**Torque Value:** 45 n•m [ 33 ft-lb ]




**⚠ CAUTION ⚠**

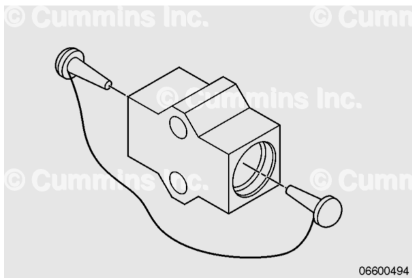
**A small amount of dirt and debris can be very harmful to the injectors and the cone seats on the injector high-pressure supply connections. Extra care is required to keep the fuel connections clean during removal and installation. See Fuel System heading in Procedure 204-008 (General Cleaning Instructions).**



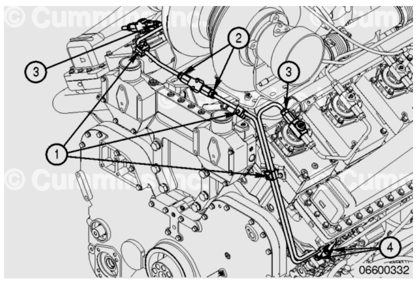

With the protective plugs still installed on the injector ports, clean the connection block ports with contact cleaner, Part Number 3824510, or equivalent.

Make sure the plugs remain wedged in place.

Remove the protective plugs from the ports in the connection block and discard the plugs.




Lubricate the pre-sliced grommets with clean engine oil on the outside and inside diameters.




Install the grommets on the outer wall of the injector supply line which connects the fuel pump to the first injector.

Remove the yellow threaded caps from the ends of the injector supply line and discard the caps.

Install new o-rings on each cone end of the injector supply line.

Insert the cone seat tips of both ends of the injector (3) and pump outlet (2).

Be careful **not** to damage the cone seats during installation.

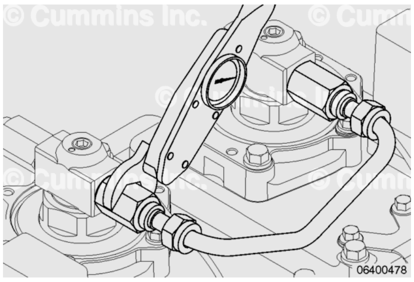
Start the first thread of both connector nuts on both ends of the injector supply line by hand. **Only** start the first thread for one connector, then start the first thread for the second connector. Be careful to properly engage the first thread to reduce the possibility of cross-threading.

Thread the connector nuts into the injectors until finger-tight. Both nuts **must** be snug before tightening either end.

Check for alignment or installation issues before final tightening.

Tighten the six connector nuts (two at the fuel pump, one at the first injector of each bank, and two at the connection block) with a suitable M27 [1-1/16 in] crow's foot and torque wrench.

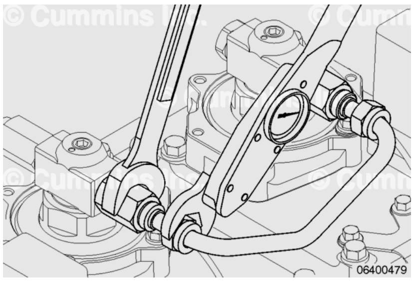
**Torque Value:** 45 n•m [ 33 ft-lb ]



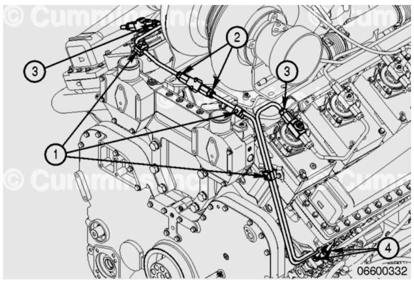

Install the rectangular grommet nuts with a suitable M24 [15/16 in] crow's foot and torque wrench.

Support the connector nut with a wrench while tightening the rectangular grommet nut.

**Torque Value:** 45 n•m [ 33 ft-lb ]



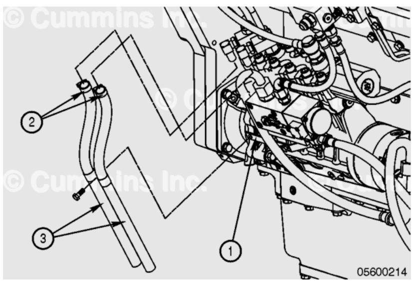

Attach the three clamps (1) to secure the injector supply lines to the engine.




Install both clamps (2) on both fuel vent hoses (3).

Install the fuel vent hoses onto the unions on the fuel pump. Tighten both clamps.

Install the capscrew and p-clips that secure the fuel vent hoses (3) to the bracket (1).



# Finishing Steps

## with Electronically Actuated Injector

- It is **not** necessary to vent air from the high-pressure fuel system before starting the engine. Cranking the engine will prime the fuel system.
- Perform a leak test.

# Leak Test

## with Electronically Actuated Injector

**⚠ WARNING ⚠**

**The fuel pump, high-pressure fuel lines, and fuel rail contain very high-pressure fuel. To reduce the possibility of personal injury, never loosen any fittings while the engine is running.**

**⚠ WARNING ⚠**

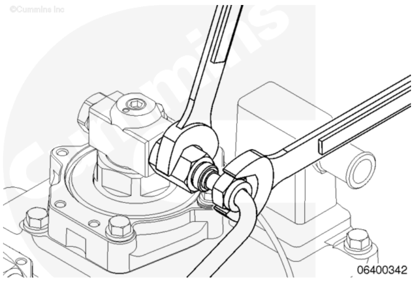
**Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.**

**⚠ WARNING ⚠**

**The pressure of the fuel in the line can penetrate the skin and cause serious personal injury. Wear gloves and protective clothing when working on the fuel system.**

**⚠ WARNING ⚠**

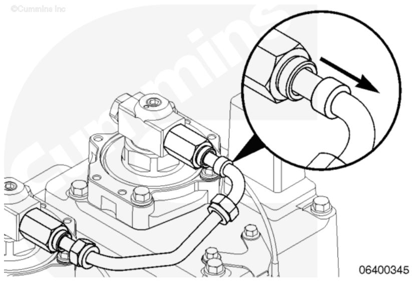
**Pressure within the high-pressure fuel system must never be measured using a mechanical gauge. Fuel pressure values of over 1724 bar [25,000 psi] are possible. Under high pressure,**



**a mechanical gauge can fail causing a high-pressure fuel leak resulting in personal injury and property damage.**

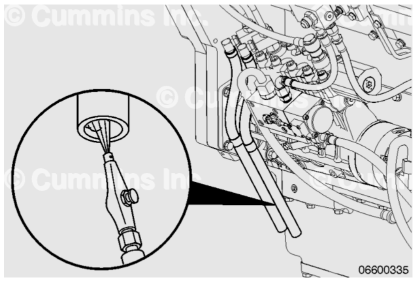
While supporting the high-pressure connector nut with a wrench, loosen the rectangular grommet nuts of the injector supply line located at the rear of the engine. Do **not** loosen the high-pressure connector nut.

Slide the rectangular grommets away from the connector fittings.




**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**




Place a rag over the injector to prevent accidental spraying of fuel on the engine.

Blow compressed air into the vent hose located on the side of the fuel pump to remove any fuel and debris from the outer wall of the injector supply lines.

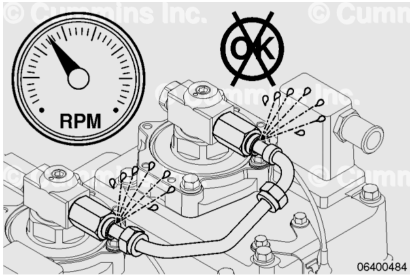
**⚠ WARNING ⚠**

**The Fuel System Leakage Test operates the fuel system at a high fuel pressure. When running the Fuel System Leakage Test through INSITE™ electronic service tool, wear safety glasses and stay clear of the engine when running the test.**


Connect INSITE™ electronic service tool and operate the Fuel System Leakage Test. See INSITE™ electronic service tool manual for instructions.

If leakage occurs at the connection(s), stop the Fuel System Leakage Test and shut off the engine.

Check to make sure the connecting nut(s) are properly torqued by loosening a half turn, then torque again. Use a suitable M27 [1-1/16 inch] crow's foot and torque wrench.



**Torque Value:** 45 n•m [ 33 ft-lb ]

Perform the Fuel System Leakage Test again and see if the leak has stopped. If the leak has stopped at both connections, install the grommets and the rectangular grommet nuts to complete the repair.

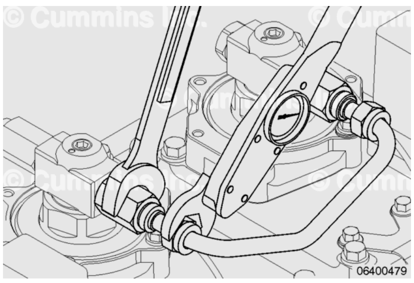
If leakage still occurs at either end of the injector supply line, remove the injector supply line and check for damage. Inspect the end(s) of the injector supply line for any signs of burrs, material rounding, or cracking. Also, check the injector fitting mating surfaces for damage. Replace damaged components as necessary.

If no signs of damage are noted, the leakage is most likely caused by misalignment during installation. Install the injector supply line again, making sure to seat the sealing cone, and tighten each connector fitting to the correct torque. Use a suitable M27 [1-1/16 inch] crow's foot and torque wrench.

**Torque Value:** 45 n•m [ 33 ft-lb ]

Perform the Fuel System Leakage Test again and check for leaks. If leakage still occurs, replace the leaking injector supply line.

If no leakage is detected, connect the rectangular grommet nut, use a suitable M24 [15/16 inch] crow's foot and torque wrench.



Support the connector nut with a wrench while tightening the rectangular grommet nut.

**Torque Value:** 45 n•m [ 33 ft-lb ]

Repeat the process for the next injector supply line. Continue with the leak test until all injector supply lines have been confirmed to have no leaks.